



RIVER FLOW 2026

THE 13TH INTERNATIONAL CONFERENCE
ON FLUVIAL HYDRAULICS

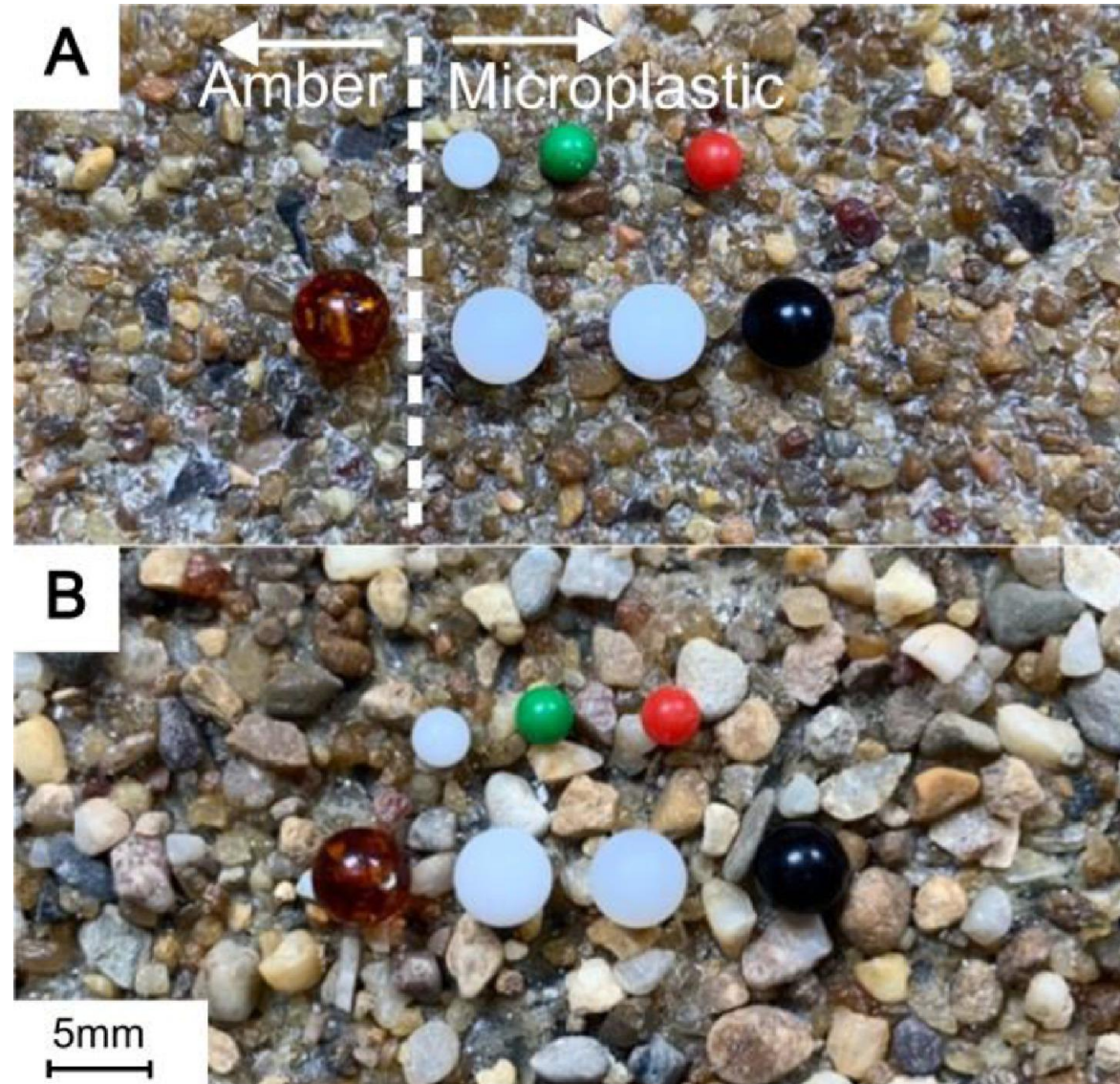
THESSALONIKI, GREECE. JUNE 30 – JULY 4, 2026

SIMPLIFYING RIVER PLASTICS

GEOMETRIC AND PHYSICAL
PROPERTIES OF THE MOST PERSISTENT LITTER ITEMS

Daniel Rebai, James Lofty, Mário J. Franca, Daniel Valero

Plastics are not spheres!



From Lofty et al, 2023

Objectives

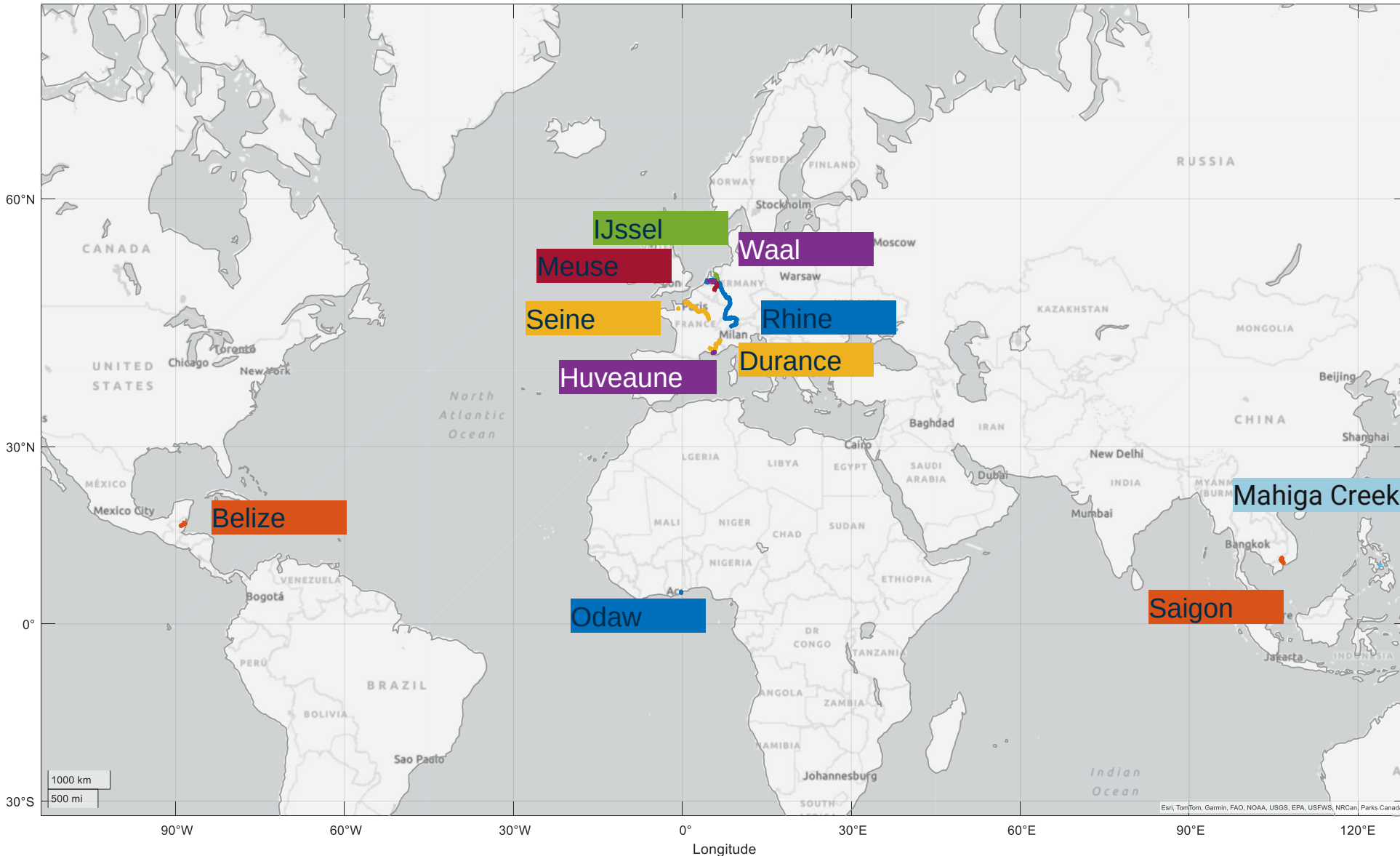
We need to understand

- Most common plastic found in riverine environments
- Their properties affecting transport

Why?

- Design of experiment
- Input for numerical codes
- Cleaning up strategies

Dataset



Sources:

- van Emmerik et al. (2020)
- Ballerini et al. (2022)
- Tramoy et al. (2019)
- de Lange et al. (2023)
- Oswald et al. (2023)
- Oswald et al. (2025)
- Vriend et al. (2023)
- Pinto et al. (2024)
- Tramoy et al. (2022)
- Silburn et al. (2023)
- Tasseron et al. (2024)
- Nguyen and Bui (2023)
- van Emmerik et al. (2018)
- Bardenas et al. (2023)

Dataset

River-OSPAR PROTOCOL

PLASTIC - hard		PLASTIC - soft		GLASS	
ID	NAME	ID	NAME	ID	NAME
4.1	Beverage bottles > 0.5 litre	4.3	Wrappers from beverage bottles	91	Glass bottles, jars and fragments
4.2	Beverage bottles < 0.5 litre	19	Candy, snack and crisps packaging	92	Lamps and fluorescent tubes
5	Cleaning products	24	Mesh bags (e.g., for onions/fruit)	93	Other glass
6	Hard plastic food packaging	25	Household gloves (soft plastic)		
8	Motor oil containers < 50 cm	31	Rope diameter > 1 cm		
9	Motor oil containers > 50 cm	32	Rope and cord diameter < 1 cm		
10	Jerry cans	39	Plastic bands and tie-wraps		
11	Caulking guns	39.1	Adhesive tape or fragments of it		
13	Crates or parts of them	40	Industrial packaging materials		
14	Car parts	46.2	Soft plastic pieces/films 2.5-50 cm		
15	Plastic bottle caps and lids	47.2	Soft plastic pieces/films > 50 cm		
16	Lighters	117.2	Soft plastic pieces/films 0.5-2.5 cm		
17	Writing utensils (e.g., pens)				
19.1	Plastic lollipop sticks				
20	Toys				
21	Plastic cups or parts of them				
21.2	Foam cups or parts of them				
22	Cutlery				
22.1	Straws				
22.2	Stirring sticks				
22.3	Plates				
35	Fishing gear				
35.1	Fishing thread (plastic nylon strands)				
36	Snap-off sticks				
38	Buckets or parts of them				
38.1	Flower/plant pots or planting trays				
42	Helmets				
43	Gun cartridges and shells				
43.1	Fireworks or remnants of them				
46.1	Hard plastic pieces 2.5-50 cm				
47.1	Hard plastic pieces > 50 cm				
48	Other plastics				
48.1	Biofilm/water filters				
63	Cigarette packaging				
64	Cigarette filters				
113	Professional gloves (thicker plastic)				
117.1	Hard plastic pieces 0.5-2.5 cm				

PAPER	
ID	NAME
60	Bags/sacks
61	Cardboard
62.1	Beverage cartons
65	Cardboard cups
66	Newspapers/magazines
67	Other paper
67.1	Pieces of paper 0-50 cm

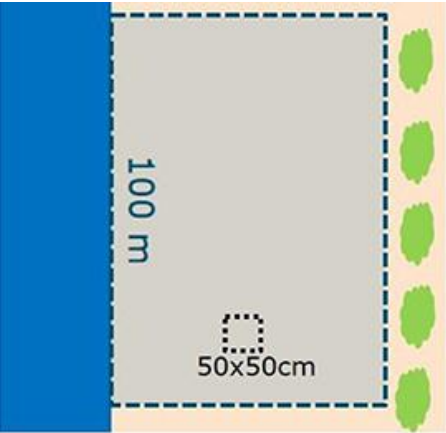
PAPER	
ID	NAME
49	Balloons or remnants of balloons
52	Tires (e.g., car/bicycle)
53	Other rubber

PLASTIC - foam	
ID	NAME
6.1	Polystyrene foam food packaging
46.3	Foam plastic pieces 2.5-50cm
47.3	Foam plastic pieces >50 cm
117.3	Foam plastic pieces 0.5-2.5 cm

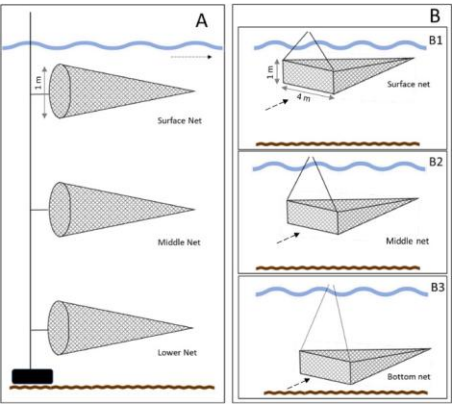
TEXTILES	
ID	NAME
54	Clothing
55	Carpets
57	Shoes, boots, and slippers
59	Other textiles

WOOD	
ID	NAME
68	Corks
69	Pallets
72	Ice cream sticks
73	Paintbrushes
74	Other wood < 50 cm
75	Other wood > 50 cm

METAL	
ID	NAME
76	Aerosol cans
77	Glass bottle caps (e.g., beer caps)
78	Beverage cans
79	Electrical wires
80	Fishing sinkers
81	Aluminum foil and packaging
81.1	Capsules (e.g., coffee/hot chocolate)
82	Food cans
83	Industrial scrap metal
84	Oil drums
86	Paint cans
88	Fencing wire, barbed wire
89	Metal pieces < 50 cm
90	Metal pieces > 50 cm
120	Disposable BBQs



van Emmerik et al. 2020



Oswald et al. 2023

Ospar	Items
46.2	465

De Lange et al. 2023

Ospar	L1 (cm)	L2 (cm)	M (g)
46.2	12.5	5.6	8.9

So far, no information about the volume V and the shortest dimension L3!

Synthetic dataset

- We generated a synthetic dataset using copulas.
- We assume a triangular distribution for density, with ranges defined by the polymer:

$$\rho \sim Tri(\rho_{min}, \rho_{max})$$

- We model volume as:

$$V = kL_1L_2L_3$$

- We compute the shortest dimension as:

$$L_3 = \frac{M}{\rho k L_1 L_2}$$

Synthetic dataset

- We generated a synthetic dataset using copulas.
- We assume a triangular distribution for density, with ranges defined by the polymer:

$$\rho \sim \text{Tri}(\rho_{\min}, \rho_{\max})$$

- We model volume as:

$$V = kL_1L_2L_3$$

- We compute the shortest dimension as:

$$L_3 = \frac{M}{\rho k L_1 L_2}$$

Material category	Common usage	Density range (g/cm ³)
Soft polyolefin (PO soft)	Flexible plastics: plastic bags, films, foils and wrappings	0.83–0.98
Hard polyolefin (PO hard)	Rigid plastics: bottle caps, lighters and hard containers	0.83–0.92
Polystyrene (PS)	Plastic cutlery, food containers, straws and cups	1.04–1.10
Expanded polystyrene (EPS)	Foams, packaging, takeaway containers	0.01–0.04
Polyethylene terephthalate (PET)	Plastic bottles	1.35–1.45
Multilayer	Cigarettes, crisp packets, candy bar wrappers, juice or milk cartons	0.89–1.45
Other plastics	Plastic items not described in the River-OSPAR litter index	0.89–1.45
Glass	Glass bottles and jars	2.20–2.80
Metal	Beverage cans, glass bottle caps, scrap metal	2.70–8.00
Rubber	Tyres, balloons and condoms	1.10–1.20
Paper	Cardboard packaging, toilet paper, newspaper and cups	0.60–1.20
Textiles	Clothing, carpets, fabrics	1.30–1.80
Wood	Pallets, corks, ice cream sticks	0.30–1.00

Synthetic dataset

- We generated a synthetic dataset using copulas.
- We assume a triangular distribution for density, with ranges defined by the polymer:

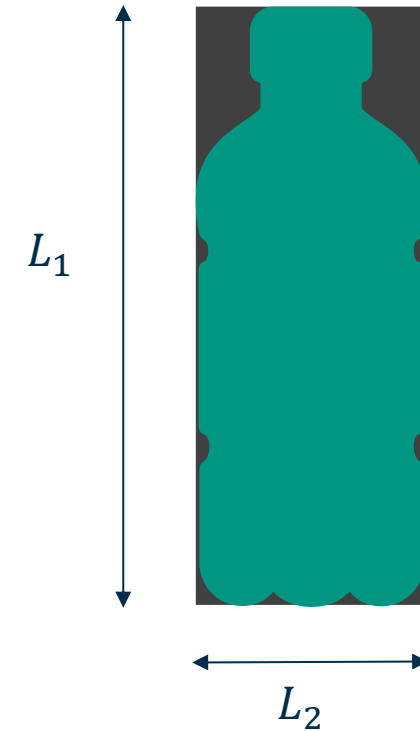
$$\rho \sim \text{Tri}(\rho_{\min}, \rho_{\max})$$

- We model volume as:

$$V = kL_1L_2L_3$$

- We compute the shortest dimension as:

$$L_3 = \frac{M}{\rho k L_1 L_2}$$



$$k = \frac{A}{L_1 L_2} \sim U(0,1)$$

Synthetic dataset

- We generated a synthetic dataset using copulas.
- We assume a triangular distribution for density, with ranges defined by the polymer:

$$\rho \sim Tri(\rho_{min}, \rho_{max})$$

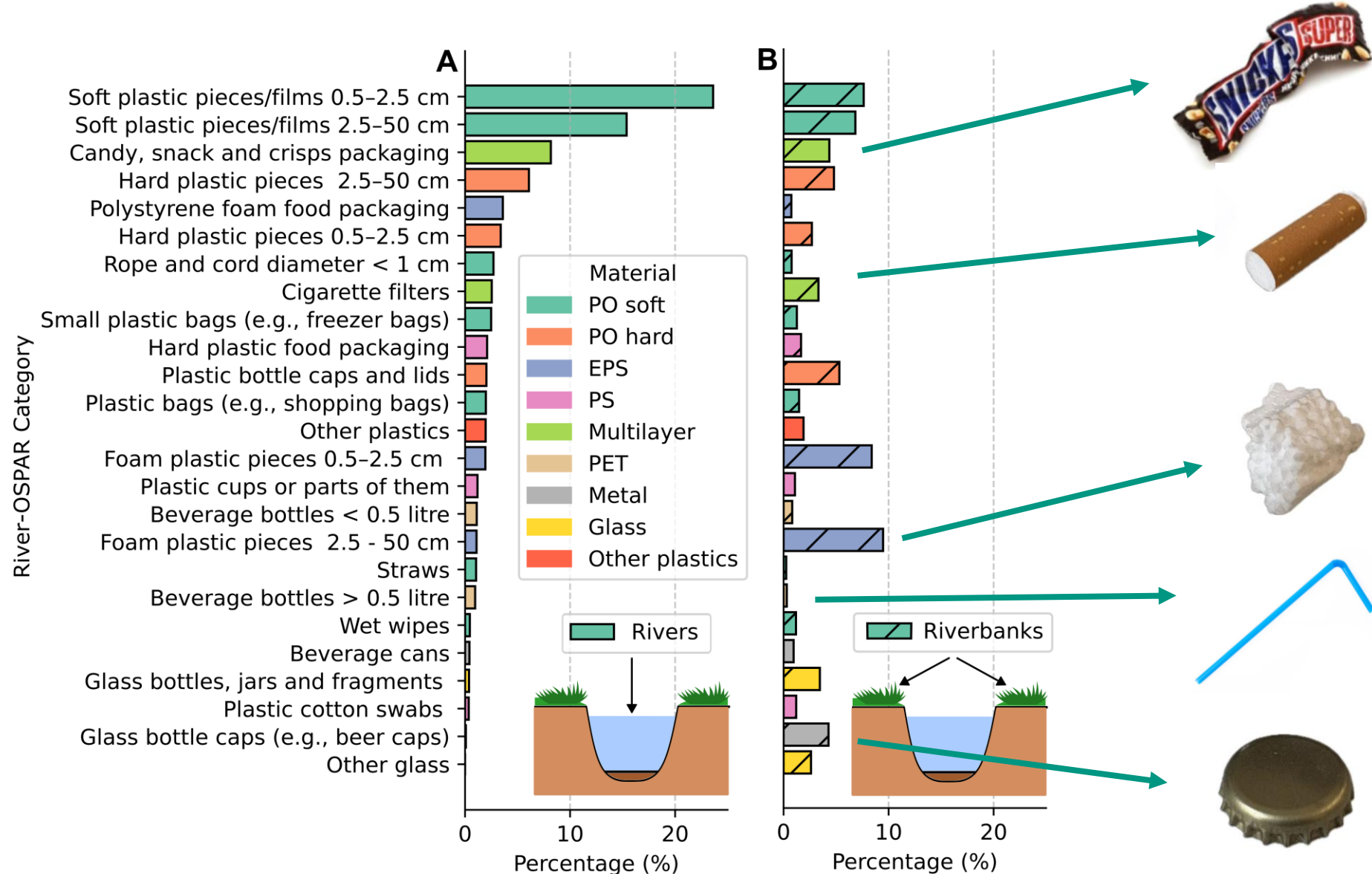
- We model volume as:

$$V = kL_1L_2L_3$$

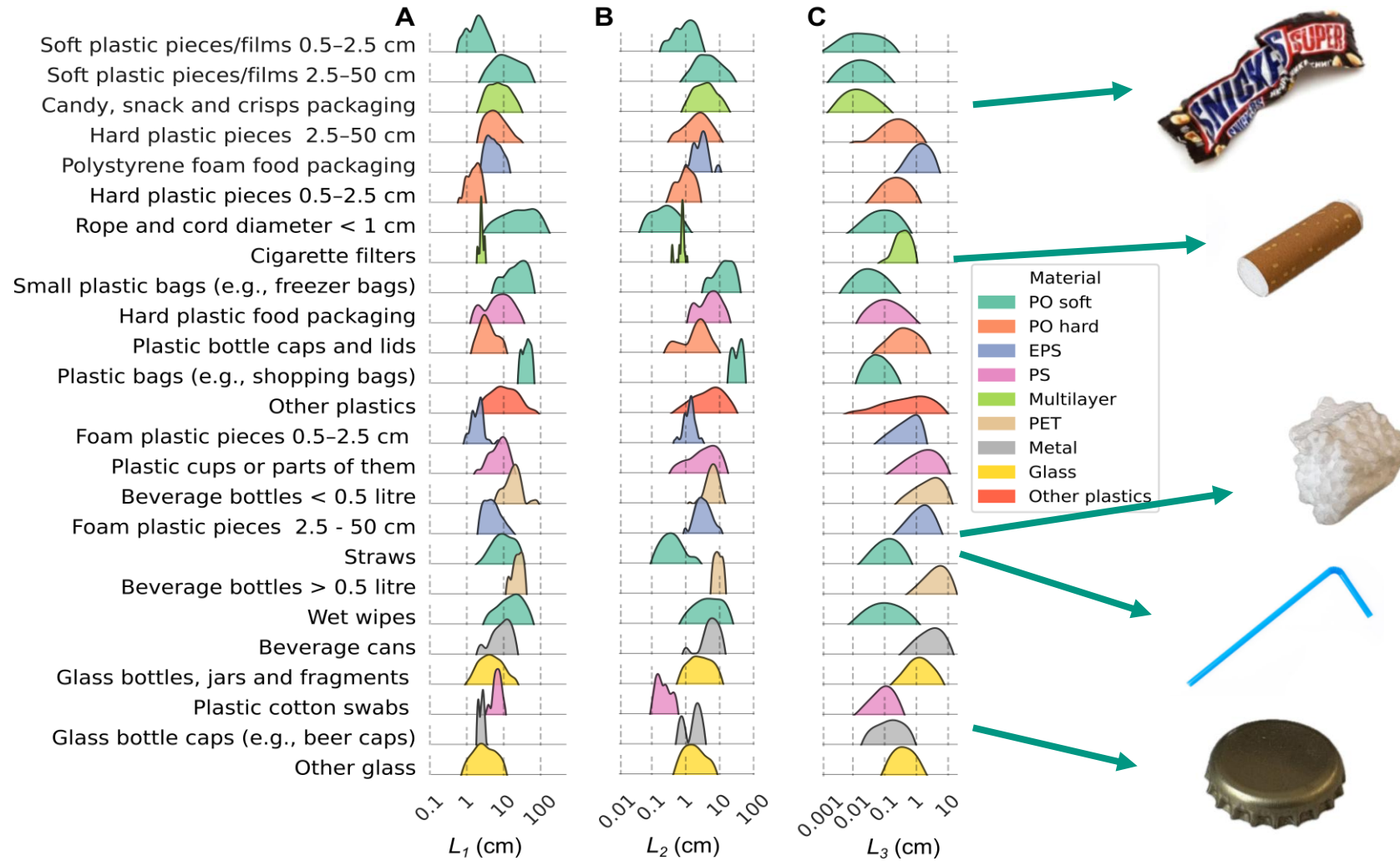
- We compute the shortest dimension as:

$$L_3 = \frac{M}{\rho k L_1 L_2}$$

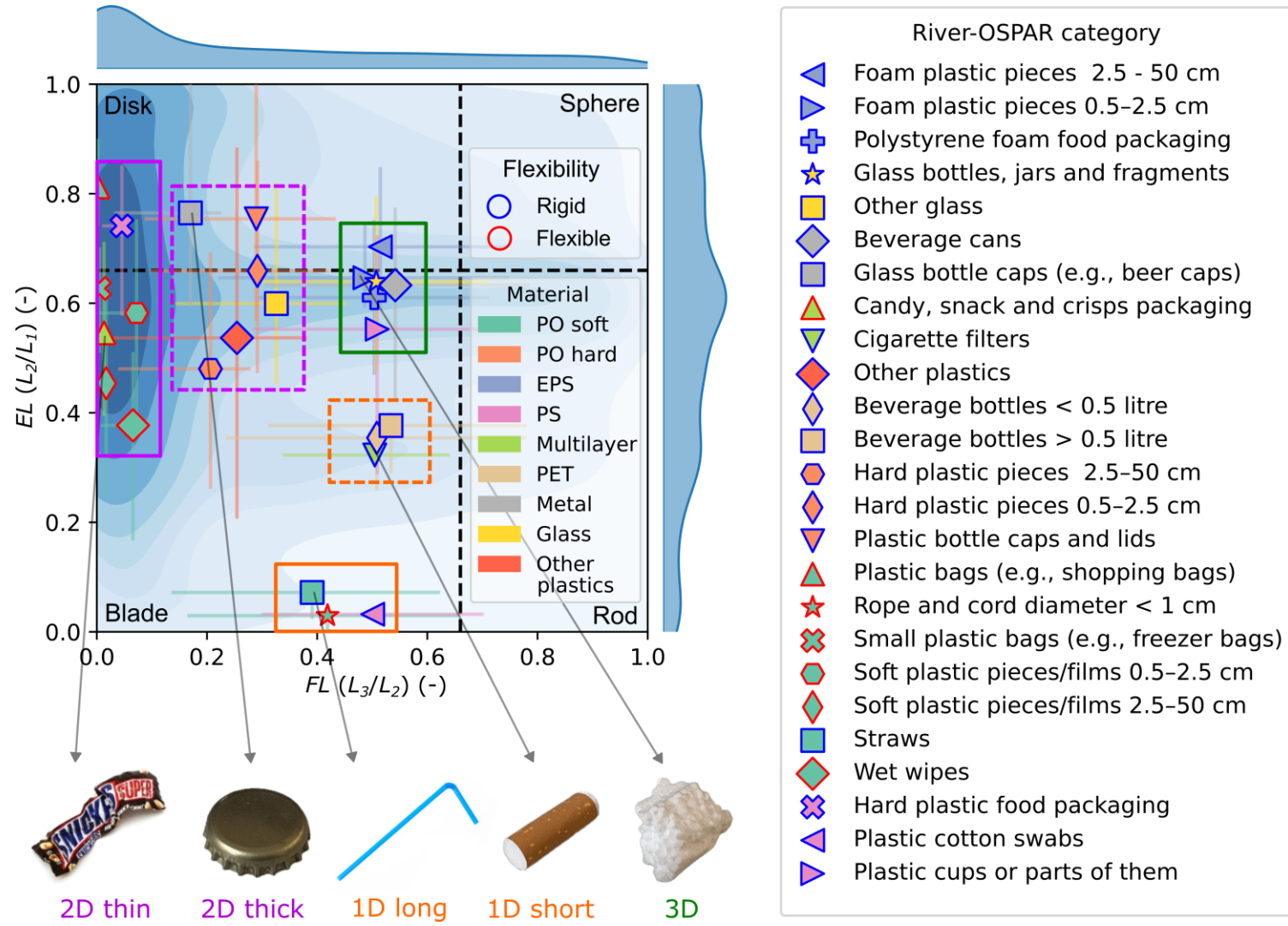
Most common litter in riverine environment



Size of the most common litter



Shape of the most common macrolitter



Conclusion

We now know:

- Most common macrolitter
- Sizes
- Shape
- Material and density



pubs.acs.org/est

Article

The Shapes and Sizes of Macroplastics and Other Litter in Rivers

James Lofty, Daniel Rebai,* Daniel Valero, and Mário J. Franca



Cite This: <https://doi.org/10.1021/acs.est.5c13126>



Read Online

ACCESS |



Metrics & More



Article Recommendations



Supporting Information



Deutsche
Forschungsgemeinschaft